MAT 3007 OPTIMIZATION
CUHK-SHENZHEN • SDS
MINGHUA CHEN • XIAO LI • YUANG CHEN

MIDTERM EXAMINATION 25 FALL

Nov. 1, 2025

Question	Points	Score
True/False	15	
The Simplex Method	17	
Duality	19	
Sensitivity Analysis	23	
Modeling	12	
Proving von Neumann's Minimax Theorem via LP Duality	14	
Total:	100	

- Please write down your **student ID** on the **answer paper**.
- Please justify your answers except Question 1.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

Question 1 [15 points]: True/False

State whether each of the following statements is *True* or *False*. For each part, only your answer, which should be one of True or False, will be graded. Explanations are not required and will not be read.

(a) [3 points] If an LP has an optimal solution, the simplex method will eventually find a BFS optimal solution, while the interior point method finds a non-BFS optimal solution.

Solution: False. This LP can only have a unique optimal solution, which must be a BFS.

(b) [3 points] The feasible region of any LP (assume the LP is feasible) must be a polyhedron.

Solution: True.

(c) [3 points] For any feasible primal and dual points $\boldsymbol{x}, \boldsymbol{y}$, we have that the primal objective function value of $\boldsymbol{x}$ is always lager than the dual objective function value of $\boldsymbol{y}$.

Solution: False. It depends on the specific min or max.

(d) [3 points] A standard form LP can have infinitely many optimal solutions.

Solution: True.

(e) [3 points] Even if the reduced costs have negative value, the current BFS may still be an optimal solution.

Solution: True. Can happen in the degenerate case.

Question 2 [17 points]: The Simplex Method

Consider the following LP problem:

$$\begin{aligned} \underset{x_1,x_2,x_3,x_4,x_5,x_6}{\text{minimize}} \quad & -x_1 - 3x_3 + 3x_4 - 4x_5 \\ \text{subject to} \quad & -x_3 + x_4 + x_5 + x_6 = 1 \\ & x_1 - x_2 + x_3 - x_4 = 2 \\ & 2x_2 + 3x_3 - 3x_4 - x_5 = 2 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \end{aligned}$$

(a) [10 points] Consider the BFS $\boldsymbol{x}$ with basis $B = \{1, 3, 6\}$. Implementing one iteration of the simplex method at $\boldsymbol{x}$ through the following steps: 1) compute and list an index j to enter the basis using the smallest index rule; 2) then, compute and list the corresponding j-th basic direction $\boldsymbol{d}$ and stepsize θ^*; 3) finally, compute and list $\boldsymbol{y}$ using the updating rule $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$, and specify the basis B and non-basic part N of $\boldsymbol{y}$.
You are given that

$$\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0, & 1, & -\frac{1}{3} \\ 0, & 0, & \frac{1}{3} \\ 1, & 0, & \frac{1}{3} \end{bmatrix}$$

Note: You do not need to use simplex tableau in this question.

Solution: We first need to compute the reduced costs $\bar{\boldsymbol{c}}$ as

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A} = \left[0, \tfrac{1}{3}, 0, 0, -\tfrac{14}{3}, 0\right]^\top. \quad \text{(2pts)}$$

The only choice is $j = 5$ to enter the basis (1pts). Correspondingly, we can compute the j-th basic direction

$$\boldsymbol{d} = [-\boldsymbol{A}_B^{-1} \boldsymbol{A}_j, 0, \ldots, 1, \ldots, 0] = \left[-\tfrac{1}{3}, 0, \tfrac{1}{3}, 0, 1, -\tfrac{2}{3}\right]^\top$$

(3pts). 1pts for the formula. 2pts for the final answer of $\boldsymbol{d}$. If their order of the numbers are wrong, they cannot get the score.

To compute θ^*, we need to first compute the current BFS $\boldsymbol{x}$. $\boldsymbol{x}_B = \boldsymbol{A}_B^{-1} b = \left[\frac{4}{3}, \frac{2}{3}, \frac{5}{3}\right]^\top$. Hence, $\boldsymbol{x} = \left[\frac{4}{3}, 0, \frac{2}{3}, 0, 0, \frac{5}{3}\right]^\top$ (1pts). The stepsize θ^* is then given by (leaving index is $B(\ell) = 6$)

$$\theta^* = \min_{i \in B, d_i < 0} -\frac{x_i}{d_i} = \frac{5}{2}. \quad \text{(1pts)}$$

Therefore,

$$\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d} = \left[\tfrac{1}{2}, 0, \tfrac{3}{2}, 0, \tfrac{5}{2}, 0\right]^\top. \quad \text{(1pts)}$$

The new basis $B = \{1, 3, 5\}$ and non-basic part $N = \{2, 4, 6\}$ (1pts).

(b) [7 points] Starting with the following simplex tableau, continue to solve the LP completely. For each step, clearly mark the current basis, the current BFS, and the corresponding objective value, the entering index and the leaving index.

B	0	0	0	0	-5	-1/2	5/2
1	1	0	0	0	2	5/2	11/2
2	0	1	0	0	1	3/2	5/2
4	0	0	-1	1	1	1	1

Solution: The entering index is $j = 5$ and the leaving index is $B(\ell) = 4$ (1pts). After the pivot updates, we obtain the new tableau

B	0	0	-5	5	0	9/2	15/2
1	1	0	2	-2	0	1/2	7/2
2	0	1	1	-1	0	1/2	3/2
5	0	0	-1	1	1	1	1

The current basis is $B = \{1, 2, 5\}$, the current BFS is $\boldsymbol{x} = [7/2, 3/2, 0, 0, 1, 0]^\top$, the corresponding objective value is $-15/2$, and the current BFS is not optimal due to negative reduced cost and non-degenerate BFS (3pts, including the tableau). The entering index is $j = 3$ and the leaving index is $B(\ell) = 2$ (1pts). After the pivot updates, we obtain the new tableau

B	0	5	0	0	0	7	15
1	1	-2	0	0	0	$-1/2$	1/2
3	0	1	1	-1	0	1/2	3/2
5	0	1	0	0	1	3/2	5/2

The current basis $B = \{1, 3, 5\}$ is optimal because all reduced costs are nonnegative (1pts). The corresponding optimal solution is $\boldsymbol{x} = [\frac{1}{2}, 0, \frac{3}{2}, 0, \frac{5}{2}, 0]^\top$ and the optimal objective value is -15 (2pts).

Question 3 [19 points]: Duality

Consider the following linear programming problem:

$$\begin{aligned} Z^* = \text{minimize} \quad & x_1 + x_2 + \ldots + x_{2025} \\ \text{subject to} \quad & x_1 + 2x_2 + \cdots + 2025x_{2025} \quad \geq 2026 \\ & 2025x_1 + 2024x_2 + \cdots + x_{2025} \quad \geq 2026 \\ & x_1, x_2, \ldots, x_{2025} \text{ free.} \end{aligned}$$

(a) [7 points] Derive the dual problem.

Note: You do not need to list all the 2025 dual constraints one by one, listing the first two and the last one to indicate the trend is sufficient.

Solution: The dual problem is given by

$$\begin{aligned} \max \quad & 20206y_1 + 2026y_2 \quad \text{(1pts)} \\ \text{s.t.} \quad & y_1 + 2025y_2 = 1 \\ & 2y_1 + 2024y_2 = 1 \\ & \vdots \\ & 2025y_1 + y_2 = 1 \quad \text{(all constraints: 4pts)} \\ & y_1, y_2 \geq 0. \quad \text{(2pts)} \end{aligned}$$

Or it's equivalent to

$$\begin{aligned} \max \quad & 20206y_1 + 2026y_2 \\ \text{s.t.} \quad & jy_1 + (2026 - j)y_2 = 1, \quad j = 1, \ldots, 2025 \\ & y_1, y_2 \geq 0. \end{aligned}$$

(b) [7 points] Find Z^* using the derived dual problem.

Solution: From the first and last equality constraints in dual:

$$\begin{cases} y_1 + 2025y_2 = 1, \\ 2025y_1 + y_2 = 1. \end{cases}$$

Solving the two-equation system gives

$$y_1 = y_2 = \frac{1}{2026}. \quad \text{(2pts)}$$

It's easy to see $y_1 = y_2 = \frac{1}{2026}$ is the only feasible solution to the dual. Thus, the optimal dual variables are $y_1^* = y_2^* = \frac{1}{2026}$ and the optimal dual objective value is

$$2026(y_1^* + y_2^*) = 2026 \times \frac{2}{2026} = 2. \quad \text{(3pts)}$$

By strong duality, $Z^* = 2$. (2pts)

(c) [5 points] Show that the two primal constraints must be active (i.e., equality holds) at the optimal solution. Then, construct one explicit primal optimal solution.

Solution: The complementary slackness conditions for the primal problem are:

$$\begin{cases} y_1^*\left(x_1^* + 2x_2^* + \cdots + 2025x_{2025}^* - 2026\right) = 0, & \text{(1pt)} \\ y_2^*\left(2025x_1^* + 2024x_2^* + \cdots + x_{2025}^* - 2026\right) = 0. & \text{(1pt)} \end{cases}$$

At optimality, we have $y_1^* = y_2^* = \frac{1}{2026} > 0$, so both terms in parentheses must equal zero. Hence, both primal constraints must be binding(active) at optimality:

$$\begin{cases} x_1^* + 2x_2^* + \cdots + 2025x_{2025}^* = 2026, \\ 2025x_1^* + 2024x_2^* + \cdots + x_{2025}^* = 2026. & \text{(1pt)} \end{cases}$$

Students can earn the above 1pt once they mention that both terms in parentheses must equal zero.

To construct an explicit primal optimal solution, observe that the two equations are symmetric. We can set

$$x_1^* = x_{2025}^* = 1, \quad \text{and } x_j^* = 0 \text{ for } j = 2, \ldots, 2024. \quad \text{(2pts)}$$

(**Note:** Any other solution satisfying the above two equations is correct.)

Question 4 [23 points]: Sensitivity Analysis

A factory makes three products: P1, P2, P3. Decision variables (units to produce):

- x_1: units of product P1.
- x_2: units of product P2.
- x_3: units of product P3.

Profit per unit:

- P1 = 3$, P2 = 2$, P3 = 4$.

Resources and budgets:

- Machine hours (max 31 hours): every unit of P1 uses 2 hours, every unit of P2 uses 1 hour, every unit of P3 uses 3 hours.
- Labor hours (max 23 hours): every unit of P1 uses 1 hour, every unit of P2 uses 2 hours, every unit of P3 uses 1 hour.
- Raw material units (max 35 units): every unit of P1 uses 3 units of materials, every unit of P2 uses 2 units of materials, every unit of P3 uses 2 units of materials.

LP formulation of this production planning problem is given maximizing the total profit:

$$\begin{array}{lll} \text{maximize} & 3x_1 + 2x_2 + 4x_3 & \\ \text{subject to} & 2x_1 + x_2 + 3x_3 \le 31 & \text{(Machine)} \\ & x_1 + 2x_2 + x_3 \le 23 & \text{(Labor)} \\ & 3x_1 + 2x_2 + 2x_3 \le 35 & \text{(Material)} \\ & x_1,\, x_2,\, x_3 \ge 0\,. & \end{array}$$

The following Table 1 gives the final simplex tableau when solving the standard form of the above problem.

B	0	0	0	8/7	2/7	1/7	47
1	1	0	0	-2/7	-4/7	5/7	3
2	0	1	0	-1/7	5/7	-1/7	7
3	0	0	1	4/7	1/7	-3 /7	6

Table 1: Final Simplex Tableau for Sensitivity Analysis.

(a) [3 points] Derive the standard form of the formulated LP.

Solution:

$$\begin{aligned} \text{minimize} \quad & -3x_1 - 2x_2 - 4x_3 \\ \text{subject to} \quad & 2x_1 + x_2 + 3x_3 + s_1 = 31 \\ & x_1 + 2x_2 + x_3 + s_2 = 23 \\ & 3x_1 + 2x_2 + 2x_3 + s_3 = 35 \\ & x_1,\ x_2,\ x_3, s_1, s_2, s_3 \geq 0\,. \end{aligned}$$

1pts for the objective, 1 pts for adding the three slack variables, and 1pts for adding nonnegative constraints.

(b) [6 points] In what range can we adjust the profit of product P2 so that the optimal production planning remains optimal? Justify your answer.

Solution: In terms of sensitivity analysis, this means when the optimal basis B remains optimal. We can read from the final tableau that

$$\boldsymbol{r}_N = \left[8/7, 2/7, 1/7\right]^\top. \quad \text{(1pts)}$$

$$\boldsymbol{A}_B^{-1}\boldsymbol{A}_N = \begin{bmatrix} -2/7 & -4/7 & 5/7 \\ -1/7 & 5/7 & -1/7 \\ 4/7 & 1/7 & -3/7 \end{bmatrix} \quad \text{(1pts)}$$

As we are going to change $c_2 \in \boldsymbol{c}_B$, we have

$$\boldsymbol{e}_j = \left[0, 1, 0\right]^\top. \quad \text{(1pts)}$$

Since the problem is maximization, thus we need to ensure

$$\boldsymbol{r}_N^\top - (-\lambda)\boldsymbol{e}_j^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}_N \geq 0. \quad \text{(2pts)}$$

This give us $-2/5 \leq \lambda \leq 1$ and the range of P2's profit is $[8/5, 3]$ (1pts).
Note: Students may use λ rather than $-\lambda$ in the formula. If they use λ, then the final range of P2's profit will be important. If they have the correct range, they get the scores. If this range is wrong, 2pts need to be deducted.

(c) [6 points] In what range can we change the total machine hours so that the original optimal basis remains optimal? What is the new dual optimal solution for the standard form primal problem? Justify your answer.

Solution: From the tableau, we have

$$\boldsymbol{x}_B^* = \left[3, 7, 6\right]^\top. \quad \text{(1pts)}$$

$$\boldsymbol{A}_B^{-1} = \begin{bmatrix} -2/7 & -4/7 & 5/7 \\ -1/7 & 5/7 & -1/7 \\ 4/7 & 1/7 & -3/7 \end{bmatrix} \quad \text{(1pts)}$$

$$\boldsymbol{e}_i = \left[1, 0, 0\right]^\top. \quad \text{(1pts)}$$

Therefore, we have

$$\boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1} \boldsymbol{e}_i \geq 0. \quad \text{(1pts)}$$

This gives $-21/2 \leq \lambda \leq 21/2$ (1pts).
The dual optimal solution is

$$\boldsymbol{y}^* = \boldsymbol{A}_B^{-\top} \boldsymbol{c}_B = \left[-8/7, -2/7, -1/7\right]^\top, \quad \text{(1pts)}$$

where $\boldsymbol{c}_B$ is from the standard form.

(d) [8 points] If the company investigates a new product P4, which has profit 2$ per unit. And every unit of product P4 consumes 1 unit of machine hour, 1 unit of labor hour, and 1 unit of raw material. What is the new optimal production planning after adding P4? What is the new optimal total profit? Justify your answer.
Hint: Adding one more variable is to add one more corresponding column in the simpelx tableau.

Solution: We use x_4 to represent the units of product P4. We first compute the corresponding reduced costs of x_4 under the current optimal basis B:

$$\bar{c}_4 = c_4 - c_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_4 = -2 - c_B^\top \boldsymbol{A}_B^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = -3/7. \quad \text{(2pts)}$$

Here, $c_4 = -2$ in standard form. This means that the original basis B is no longer optimal. We compute

$$\boldsymbol{A}_B^{-1} A_4 = \begin{bmatrix} -1/7 \\ 3/7 \\ 2/7 \end{bmatrix}. \quad \text{(1pts)}$$

Then, based on the final tableau of the original problem, we form the following new tableau by adding a new forth column with $\bar{c}_4$ and $\boldsymbol{A}_B^{-1} A_4$.

B	0	0	0	-3/7	8/7	2/7	1/7	47
1	1	0	0	-1/7	-2/7	-4/7	5/7	3
2	0	1	0	3/7	-1/7	5/7	-1/7	7
3	0	0	1	2/7	4/7	1/7	-3 /7	6

1pts for forming the tableau.
Then, we continue the simplex method for one more pivoting update and get the following new optimal solution ($\bar{c}$ nonnegative)

B	0	1	0	0	1	1	0	54
1	1	1/3	0	0	-1/3	-1/3	2/3	16/3
4	0	7/3	0	1	-1/3	5/3	-1/3	49/3
3	0	-2/3	1	0	2/3	-1/3	-1/3	4/3

2pts for updating the above tableau, deduct 1pts if one part is wrong until all 2pts are deducted.
Therefore, the new optimal planning is $16/3$ units of product P1, 0 units of product P2, $4/3$ units of product P3, and $49/3$ units of product P4, and the new optimal total profit is 54 (2pts in total, deduct 1pts if one part is wrong until all 2pts are deducted).

Question 5 [12 points]: Modeling

A company produces a dairy-based food product and plans its production for the next 12 months. Initially, it has I_0 kilograms of the product in inventory. The demand in month t is D_t kilograms. The company may produce more than the demand and store the excess as inventory. However, due to spoilage, only 95% of the inventory at the end of month t remains available at the beginning of month $t+1$ (i.e., 5% of the inventory is lost).

To ensure a stable production schedule, the company requires that the maximum absolute difference in production quantities between any two months does not exceed L kilograms. The production cost in month t is C_t dollars per kilogram, and the holding cost of inventory in month t is H_t dollars per kilogram.

Formulate an optimization problem to determine the company's optimal production plan that minimizes the total cost. Then, transform the formulated problem to an LP equivalently.

Solution: Let x_t denote the amount (in kilograms) of product to produce, and I_t denote the amount (in kilograms) of product to hold in month t, $t = 1, \ldots, 12$ (2pts). Then the problem can be formulated as:

$$\begin{aligned}
\min \quad & \sum_{t=1}^{12}(C_t x_t + H_t I_t) && && \text{(2pts)} \\
\text{s.t.} \; & I_t = 0.95 I_{t-1} + x_t - D_t, && \forall t = 1, \ldots, 12, && \text{(2pts)} \\
& \max_{t,\tau=1,\ldots,12}\{|x_t - x_\tau|\} \le L, && && \text{(2pts)} \\
& x_t,\, I_t \ge 0, && \forall t = 1, \ldots, 12. && \text{(1pt)}
\end{aligned}$$

The above problem can be transformed into a linear program as:

$$\begin{aligned}
\min \quad & \sum_{t=1}^{12}(C_t x_t + H_t I_t) && && \\
\text{s.t.} \quad & I_t = 0.95 I_{t-1} + x_t - D_t, && \forall t = 1, \ldots, 12, && \\
& x_t - x_\tau \le L, && \forall t = 1, \ldots, 12,\ \tau = 1, \ldots, 12, && \text{(3pts)} \\
& x_t,\, I_t \ge 0, && \forall t = 1, \ldots, 12. &&
\end{aligned}$$

(**Note:** Other equivalent reformulations are also fine. Writing $-L \le x_t - x_\tau \le L$ is also correct. If students write the objective as $\sum_{t=1}^{12}(C_t x_t + 0.95 H_t I_t)$ and the first set of constraints as $I_1 = I_0 + x_1 - D_1$ together with $I_t = 0.95 I_{t-1} + x_t - D_t, ; \forall t = 2, \ldots, 12$, these are also considered correct.)

Question 6 [14 points]: Proving von Neumann's Minimax Theorem via LP Duality

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a real matrix. Define two sets $\Delta_n \triangleq \{\, \boldsymbol{q} \geq 0 \,|\, \sum_{i=1}^n q_i = 1 \,\} \subset \mathbb{R}^n$ and $\Delta_m \triangleq \{\, \boldsymbol{p} \geq 0 \,|\, \sum_{j=1}^m p_j = 1 \,\} \subset \mathbb{R}^m$.

In this problem, we will use the strong duality theorem of linear programming (LP) to prove the following minimax equality:

$$\max_{\boldsymbol{q}\in\Delta_n} \min_{1\leq i\leq n} (\boldsymbol{A}\boldsymbol{q})_i = \min_{\boldsymbol{p}\in\Delta_m} \max_{1\leq j\leq m} (\boldsymbol{p}^\top \boldsymbol{A})_j. \qquad \text{(Minimax Equality)}$$

Here, $(\boldsymbol{A}\boldsymbol{q})_i$ and $(\boldsymbol{p}^\top \boldsymbol{A})_j$ represent the i-th and j-th entry of the vectors $\boldsymbol{A}\boldsymbol{q}$ and $\boldsymbol{p}^\top \boldsymbol{A}$, respectively.

(a) [4 points] Derive equivalent LP formulations for the two optimization problems:

$$\max_{\boldsymbol{q}\in\Delta_n} \min_{1\leq i\leq n} (\boldsymbol{A}\boldsymbol{q})_i$$

and

$$\min_{\boldsymbol{p}\in\Delta_m} \max_{1\leq j\leq m} (\boldsymbol{p}^\top \boldsymbol{A})_j.$$

Denote these LPs as problems P1 and P2, respectively. Prove that the feasible region of problem P1 is non-empty (e.g., by identifying a feasible point) and bounded (e.g., by showing all feasible variables cannot exceed some upper bounds).

Solution: Consider the first problem: $\max_{\boldsymbol{q}\in\Delta_n} \min_{1\leq i\leq n}(A\boldsymbol{q})_i$. We introduce an auxiliary scalar variable v requiring that

$$(A\boldsymbol{q})_i \geq v, \quad i = 1, \ldots, n. \quad \text{(0.5 pts)}$$

Then the problem can be expressed as the following LP:

$$\begin{aligned} \max \quad & v, \\ \text{s.t.} \quad & A\boldsymbol{q} \geq v\mathbf{1}, \\ & \mathbf{1}^\mathsf{T}\boldsymbol{q} = 1, \\ & \boldsymbol{q} \geq 0, \\ \text{var.} \quad & v \in \mathbb{R}, \boldsymbol{q} \in \mathbb{R}^n. \end{aligned} \qquad \text{(P1 (1 pt))}$$

We denote this problem as P1.

The feasible region of the LP problem P1 is non-empty as a feasible point is $\boldsymbol{q} = \frac{1}{n}\mathbf{1}$ with a corresponding feasible v. It is clearly bounded as all non-negative q_i is no larger than 1 and v is no larger than the sum of absolute values of the entries in A. (1 pt).

Similarly, for the second problem: $\min_{\boldsymbol{p}\in\Delta_m} \max_{1\leq j\leq m}(\boldsymbol{p}^\mathsf{T}A)_j$, we also introduce a scalar variable u requiring that

$$(\boldsymbol{p}^T A)_j \leq u, \quad j = 1, \ldots, m. \quad \text{(0.5 pts)}$$

Then the equivalent LP is

$$\begin{aligned} \min \quad & u, \\ \text{s.t.} \quad & \boldsymbol{p}^T A \leq u\mathbf{1}, \\ & \mathbf{1}^\mathsf{T}\boldsymbol{p} = 1, \\ & \boldsymbol{p} \geq 0, \\ \text{var.} \quad & u \in \mathbb{R}, \boldsymbol{p} \in \mathbb{R}^m. \end{aligned} \qquad \text{(P2 (1 pt))}$$

We denote this problem as P2.

(b) [8 points] Mathematically derive the dual of problem P1, e.g., by deriving equivalent formulations of P1 to obtain its dual as discussed in slides and lecture. Show that the dual of problem P1 is exactly P2. (Note: answers without mathematical derivation steps, e.g., directly obtaining the dual by checking the primal-dual correspondence table, receive zero point.)

Solution: We first note that from (a), the feasible region of the LP problem P1 is feasible and bounded. Thus the strong duality theorem implies that the dual of P1 (albeit we do not know its formulation yet) is also feasible and have the same optimal objective value as P1. (1 pt)

Next, we derive the dual of P1, given that we already know that its dual problem has the same optimal objective value as P1.

Next, we follow the procedure in the lecture notes to mathematically derive the dual of P1, given that we know that the strong duality holds. Problem P1 can be written as

$$\max_{v\in\mathbb{R},\boldsymbol{q}\in\mathbb{R}^n,\boldsymbol{q}\geq 0} \quad v+\min_{\boldsymbol{x}\geq 0,\boldsymbol{x}\in\mathbb{R}^m}\left[\boldsymbol{x}^T\left(A\boldsymbol{q}-v\mathbf{1}\right)\right]+\min_{y\in\mathbb{R}}\left[y\left(1-\mathbf{1}^\mathsf{T}\boldsymbol{q}\right)\right], \quad \text{(2 pts)}$$

because

$$\min_{\boldsymbol{x}\geq 0,\boldsymbol{x}\in\mathbb{R}^m}\boldsymbol{x}^T\left(A\boldsymbol{q}-v\mathbf{1}\right)=\begin{cases}0, & \text{if } A\boldsymbol{q}-v\mathbf{1}\geq 0;\\ -\infty, & \text{otherwise.}\end{cases} \quad \text{(1 pt)}$$

and

$$\min_{y\in\mathbb{R}} y\left(1-\mathbf{1}^\mathsf{T}\boldsymbol{q}\right)=\begin{cases}0, & \text{if } 1-\mathbf{1}^\mathsf{T}\boldsymbol{q}=0;\\ -\infty, & \text{otherwise.}\end{cases} \quad \text{(1 pt)}$$

Exchanging the max and min in the above formulation (we can do so because we already know strong duality holds) and re-arranging the terms, we obtain another equivalent formulation as follows:

$$\min_{y\in\mathbb{R}}\min_{\boldsymbol{x}\geq 0,\boldsymbol{x}\in\mathbb{R}^m}\max_{v\in\mathbb{R},\boldsymbol{q}\in\mathbb{R}^n,\boldsymbol{q}\geq 0}\left(\boldsymbol{x}^TA-y\mathbf{1}^\mathsf{T}\right)\boldsymbol{q}+v\left(1-\boldsymbol{x}^T\mathbf{1}\right)+y. \quad \text{(1 pt)}$$

By noticing the subproblems with respect to $\boldsymbol{q}$ and v in the above formulation can be simplified based on arguments similar to those used one paragraph above (1 pt):

$$\max_{\boldsymbol{q}\geq 0,\boldsymbol{q}\in\mathbb{R}^n}\left(\boldsymbol{x}^TA-y\mathbf{1}^\mathsf{T}\right)\boldsymbol{q}=\begin{cases}0, & \text{if } \boldsymbol{x}^TA\leq y\mathbf{1}^\mathsf{T};\\ -\infty, & \text{otherwise.}\end{cases}$$

and

$$\max_{v\in\mathbb{R}} v\left(1-\boldsymbol{x}^T\mathbf{1}\right)=\begin{cases}0, & \text{if } \boldsymbol{x}^T\mathbf{1}=1;\\ -\infty, & \text{otherwise.}\end{cases}$$

Thus, the formulation can be simplified to the following one:

$$\begin{aligned}\min_{y\in\mathbb{R}}\min_{\boldsymbol{x}\in\mathbb{R}^m}\quad & y,\\ \text{s.t.}\quad & \boldsymbol{x}^TA\leq y\mathbf{1}^\mathsf{T},\\ & \boldsymbol{x}^T\mathbf{1}=1,\\ & \boldsymbol{x}\geq 0.\end{aligned} \quad \text{((1 pt))}$$

This formulation is the dual problem of LP problem P1, and it is straightforward to verify that it is the same as the LP problem P2.

(c) [2 points] Apply strong duality theorem, conclude that their optimal objective values are equal, which establishes the Minimax Equality.

Solution: First note again the LP problem P1 is feasible and bounded (we have shown it in (a)) (1 pt) (need to state this condition is met before applying strong duality theorem). Then by strong duality theorem of LP and that P2 is the dual of P1, both P1 and P2 must be feasible and bounded, and their optimal objective values are the same:

$$\max_{\boldsymbol{q}\in\Delta_n} \min_{1\le i\le n} (A\boldsymbol{q})_i = \min_{\boldsymbol{p}\in\Delta_m} \max_{1\le j\le m} (\boldsymbol{p}^\mathsf{T} A)_j \quad \text{(1 pt)}$$

This equality is precisely the *von Neumann Minimax Theorem*.